

Supplementary Material

Title: Feature Selection based on the causality relation and extension of the parental relationship. Proposal of the LUISA algorithm

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Abstract: Non-causal feature selection methods may introduce errors in assessing feature relevance because they rely on a single selection criterion, which can lead to spurious correlations. The current causality methods of feature selection consider only the features that compose the Markov Blanket of one target variable (class) as relevant. That may lead to the disregard of other relevant features.

We present a new algorithm, LUISA, based on causal relationships. This algorithm expands the Markov Blanket to include indirect ancestors for a specific target variable or class. We test the algorithm on five perspectives using synthetic and real-world datasets. The experiments show that LUISA is representative and that it performs satisfactorily under uncertainty.

In addition, the learning models demonstrated promising results when using the reduced dataset compared to the complete dataset and various feature non-causal and causal selection methods.

Real-world datasets:

Breast Cancer: breast_cancer_wisconsin.zip

Heart: statlog_heart.zip

Hepatitis: hepatitis3.zip

Divorce: divorcio1.zip

Sepsi: sepsi.zip

LUISA Algorithm

LUISA uses MMPC.R available at: <https://github.com/cran/MXM/blob/master/R/MMPC.R>

LUISA uses MXM available at: <https://github.com/cran/MXM/tree/master>

LUISA is available in: LUISA_ALL.zip

(Observation: MMPC, MXM and Datasets must be in the same path)

Learning models with complete dataset and application of the non-causal filters

[KNIME Analytics Platform](#)

Workflow: Causalidade_Luisa.knwf

Application of the MMPC algorithm to the datasets

Breast Cancer: mmpc_cancermama

Heart: mmpc_heart

Hepatitis: mmpc_hepatite

Divorce: mmpc_divorcio

Sepsi: mmpc_sepsi

Experiments_MMPC.zip

Results of global experiments

Global_experiments.zip